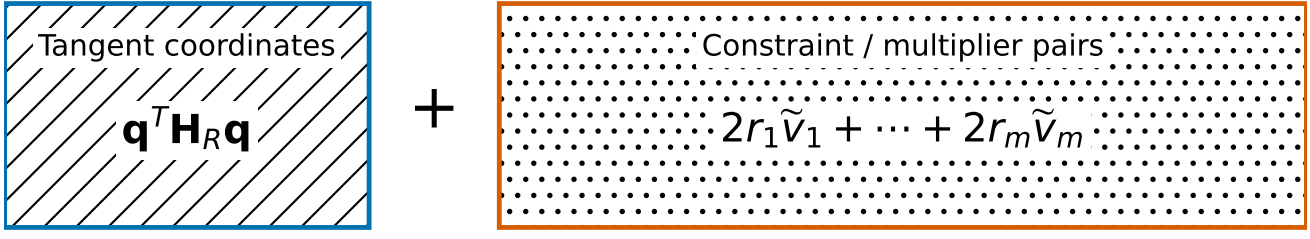

Invertible coordinates eliminate coupling: $\mathbf{v} = \tilde{\mathbf{v}} - \mathbf{G}^T\mathbf{q} - \frac{1}{2}\mathbf{H}_N\mathbf{r}$